

BLA-03 · BUILDING AI APPS

Wiring an AI App Together

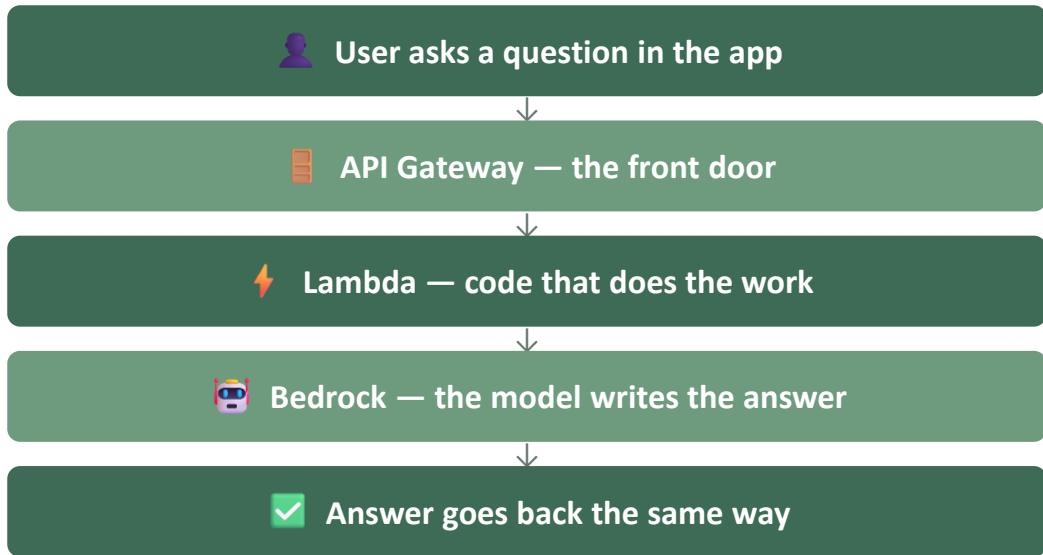
The journey of one request · AWS Lambda · Amazon API Gateway · AWS AppConfig · AWS Step Functions

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PART

02

The journey of one request



Each piece has one job. That's what makes it easy to change one part later without breaking the rest.

AWS Lambda

Small pieces of code that run only when something happens.

- No servers to manage · pay only while it runs · scales on its own
- Has a time limit — not meant for very long jobs

Pattern	Example
Wait for the answer	User asks, Lambda calls Bedrock, replies straight away
Hand it off	A file lands in S3, Lambda summarises it later
Read a stream	Processes a flow of events as they arrive

Amazon API Gateway

The front door to your app.

It handles	So your code doesn't have to
 Who's allowed in	Checks keys and logins
 How many requests	Throttles anyone sending too many
 Repeated questions	Can cache common answers
 Usage plans	Different limits for different customers

 **It only waits so long for a reply.** Long model answers need streaming, or a "we'll let you know" background pattern.

AWS AppConfig

Change how the app behaves — without redeploying it.

Switching from Model A to Model B	Steps
 Without AppConfig	Change code → test → rebuild → redeploy
 With AppConfig	Change one setting → roll out gradually → roll back if needed

 **Also useful for:** temperature, prompt versions, and switching features on or off. Anything that changes often belongs in config, not code.

AWS Step Functions

A workflow of steps, drawn as a flowchart.



Prompt chaining — handling a complaint email:



Several small, focused prompts usually beat one giant one — each step is easier to test, and you can see exactly which one went wrong.

The circuit breaker

If the model keeps failing, stop calling it for a while.

State	What happens
 Closed	Everything normal — calls go through
 Open	Too many failures — use a fallback instead
 Half-open	Try one call to test whether it has recovered



Fallbacks: a smaller model · a cached answer · "We're busy — please try again shortly"



Like the fuse box at home — it trips to stop the damage spreading.

Key takeaways

- 1 API Gateway is the front door · Lambda does the work
- 2 Long answers need streaming or a background pattern
- 3 Put anything that changes often in AppConfig, not code
- 4 Step Functions: several small prompts beat one giant one
- 5 A circuit breaker stops one failure becoming an outage

Next · Part 3: connecting it, shipping it, and where it runs